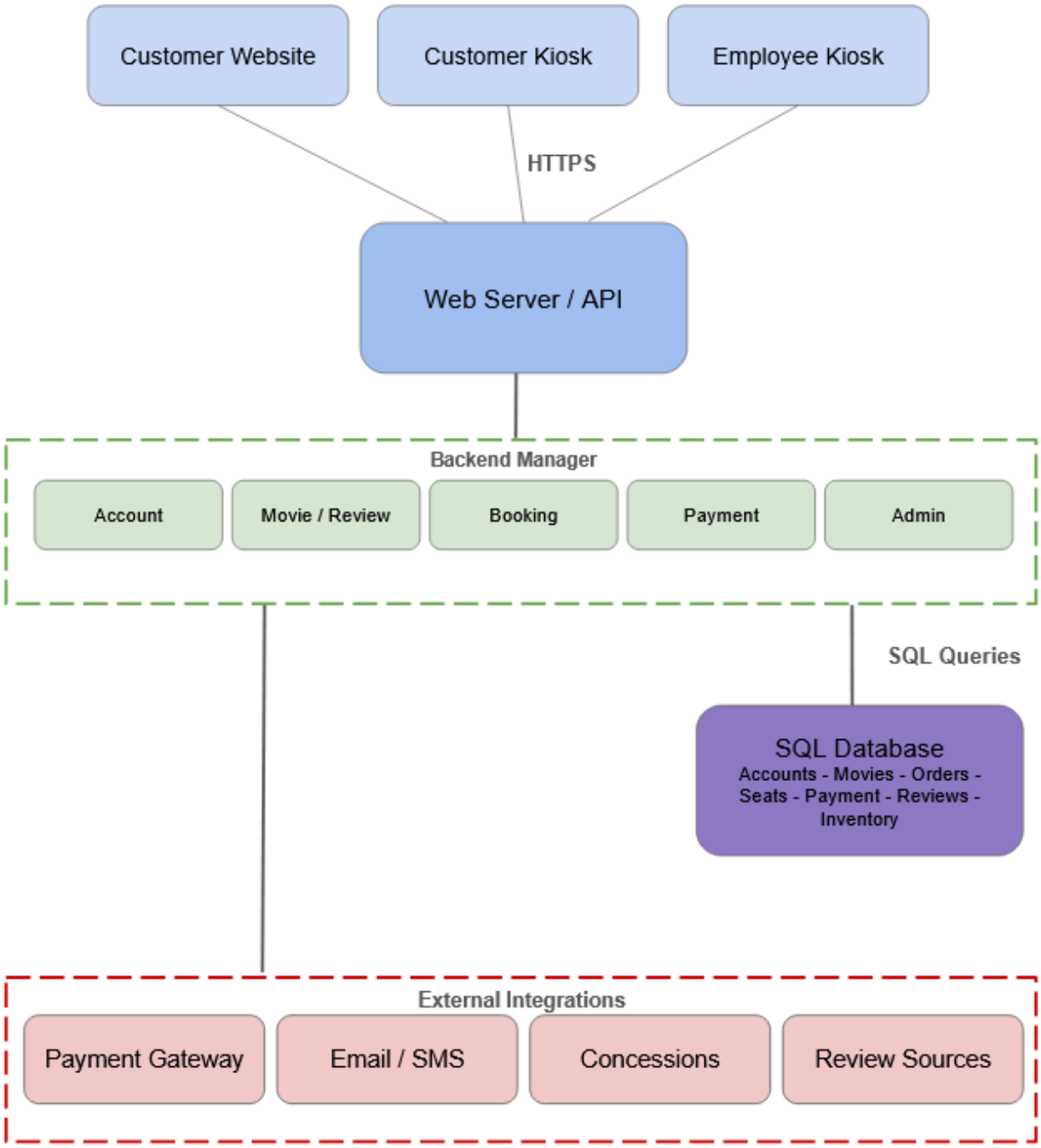
6.2.1 Software Architecture Diagram



Modifications:

- Specified the “Database” to “SQL Database”
- Labeling “SQL Queries” to show that managers read and update stored information.
- Version 1 of the SWA Diagram included the Backend Managers and External Integrations connection. “Database” description was updated.

6.4 Data Management Strategy

6.4.1 Database Technology Selection (SQL vs NoSQL)

The JJT Movies Theater Ticketing System will use a SQL database management system to store and manage information. SQL databases organize data into tables with rows, columns, primary keys, and foreign keys to identify information. Connections between tables are established though foreign keys where related information is accessed efficiently.

We choose SQL because our ticketing system uses information that is closely connected (customers, movies, showtimes, seats, and ticket orders). When purchasing a ticket the system must be able to associate that ticket with correct showtimes and live seats. A SQL database supports the transactions our system demands and the constraints that are needed to maintain accuracy and safety. It relies on complex querying, data integrity, and strict relationships. A NoSQL database would be useful when working with less structured data and stores data in a flexible way. Because our ticketing system relies on consistent relationships between records, SQL was selected because its capabilities are better suited for our systems's requirements.

One tradeoff of using SQL is that changes to the database structure may require the modification of existing tables and relationships.Despite this limitation, the benefits of maintaining consistent and organized ticketing information makes SQL the preferred choice for our system. JJT Movies' client-server architecture is the most optimized with SQL.

6.4.2 Database Organization and Table

The movie ticketing system will use a centralized SQL database because it allows all three theatre locations to update information in one place. The database will be organized into tables, with each storing specific information related to the system. Primary keys will uniquely identify records, while foreign keys will connect related tables. This allows the system to manage customer information, movie showtimes, ticket purchases, payments and concessions.

Table Name	Information Stored
Customers	Customer ID, name, email, password hash, membership ID, reward points
Memberships	Membership ID, Membership tier, monthly fee, benefits
Movies	Movie ID, Title, rating, runtime

Theatres	Theatre ID, name, address
Auditoriums	Auditorium ID, theatre ID, type, seating
Showtimes	Showtime ID, Movie ID, auditorium ID, date, time
Seats	Seat ID, Auditorium ID, row, seat number
Tickets	Ticket ID, order ID, Showtime ID, Seat ID, Ticket Type, Price
Orders	Order ID, Customer ID, Order date, total amount, Status
Payments	Payment ID, Order ID, Payment Method, Amount, Payment Status
Concessions	Concession item ID, Name, Size, Price, Available Quantity
Order Concessions	Order ID, Concession Item ID, Quantity
Reviews	Review ID, Movie ID, Customer ID, Rating, Review Text
Employees	Employee ID, Name, Employee Role

6.4.3 Database Diagram